

Understanding Large Language Models

Introduction for Data Engineers

Data Engineering Boot Camp

July 01, 2025

Agenda

Learning Outcomes

What Are Large Language Models?

Data Engineering Perspective

Historical Context and Impact

Q&A and Discussion

Next Steps

Learning Outcomes

By the end of this lesson, you will be able to:

- ▶ Define Large Language Models and explain their core principles of pattern recognition at scale
- ▶ Compare LLM data requirements to traditional data engineering scales and identify key preprocessing challenges
- ▶ Contextualize the emergence of LLMs within the broader evolution of data processing paradigms

What Are Large Language Models?

Definition: Neural networks trained on massive text datasets to understand and generate human-like language.

Core Concept

Sophisticated pattern recognition systems that learn statistical patterns of human language from vast text corpora (books, articles, code, web pages).

Key Characteristics

- ▶ **Scale:** Billions to trillions of parameters (vs. thousands in traditional ML)
- ▶ **Training Data:** Terabytes to petabytes of text
- ▶ **Emergent Abilities:** New capabilities at larger scales
- ▶ **General Purpose:** One model, many language tasks

Pattern Recognition vs. Explicit Programming

Traditional Software

- ▶ Requirements → Code → Rules → Output
- ▶ Example: Explicit SQL queries with JOINS and WHERE clauses

LLM Approach

- ▶ Data → Training → Patterns → Predictions
- ▶ Example: Learns data relationships from millions of examples

Emergent Capabilities

As LLMs scale, they develop new abilities:

- ▶ **Few-shot Learning:** Learn tasks from a few examples
- ▶ **Chain-of-Thought Reasoning:** Break down complex problems
- ▶ **Code Generation:** Write functional code from descriptions

Data Engineering Parallel: A data warehouse enables complex analytics at scale, similar to LLMs enabling new capabilities.

Key Examples

- ▶ **GPT Family (OpenAI):** GPT-3 (175B parameters), GPT-4 (1T+ parameters), trained on diverse internet text
- ▶ **Claude (Anthropic):** Focus on helpful, harmless, honest responses
- ▶ **BERT (Google):** Bidirectional context, used for search and document understanding

Text as Structured Data

Traditional Data Pipeline

- ▶ Raw Data → ETL → Data Warehouse → Analytics

LLM Data Pipeline

- ▶ Raw Text → Tokenization → Embeddings → Training → Inference

Scale Comparison

	Enterprise DW	LLM Training	Example:
Size	1–100 TB	1–10 PB	
Records/Tokens	Millions–Billions	Trillions	
Processing	Batch, overnight ETL	Months on GPUs	

GPT-3 trained on 45TB of text ($\sim$ 300B words, 600,000 books).

Data Preprocessing Challenges

1. Data Cleaning at Scale

- ▶ Deduplication across petabytes
- ▶ Quality filtering (remove low-quality/harmful content)
- ▶ Language detection for multilingual data

2. Tokenization

- ▶ Convert text to numerical tokens
- ▶ Handle languages, special characters, code

3. Data Validation

- ▶ Verify factual accuracy
- ▶ Detect and mitigate bias
- ▶ Scrub personal/sensitive data

Infrastructure Requirements

- ▶ **Storage:** Petabyte-scale distributed systems, high-throughput access
- ▶ **Compute:** Thousands of GPUs/TPUs, high-speed interconnects
- ▶ **Network:** High-bandwidth for distributed training, edge computing

Cost: Training GPT-3 costs \$4–12M in compute alone.

Evolution of Language Processing

- ▶ **1980s–1990s:** Rule-based systems, hand-coded grammar
- ▶ **2000s–2010s:** Statistical ML, bag-of-words, n-grams
- ▶ **2010s–2020s:** Neural networks, word embeddings, RNNs
- ▶ **2020s–Present:** Transformer-based LLMs, attention mechanisms

Paradigm Shift

Batch Processing

- ▶ Large chunks, scheduled jobs
- ▶ Efficient but latent

Stream Processing

- ▶ Real-time data processing
- ▶ Complex but immediate

LLM Shift: Task-specific to general-purpose, explicit to learned patterns.

Impact on Data-Driven Applications

- ▶ **Document Processing:** Summarization, search, content generation
- ▶ **Data Analysis:** NL queries, report generation, insight extraction
- ▶ **Business Intelligence:** Conversational analytics, anomaly detection
- ▶ **Customer Service:** Chatbots, email automation, sentiment analysis

Discussion Questions

1. How do LLM data volumes compare to your largest datasets?
2. Where could LLMs fit into your data pipelines?
3. What challenges do you anticipate with LLM-scale processing?

Key Takeaways

- ▶ LLMs shift from explicit programming to pattern recognition at scale
- ▶ Data requirements are unprecedented but follow DE principles
- ▶ Rapidly evolving technology impacting data-driven applications
- ▶ Understanding LLMs is essential for modern data engineers

Preparation for Next Lesson

Topics:

- ▶ LLM architecture and data flow
- ▶ Production deployment and integration strategies
- ▶ Infrastructure for enterprise LLMs

Additional Resources

- ▶ **Reading:**

- ▶ “Attention Is All You Need” (Vaswani et al., 2017)
- ▶ OpenAI GPT-3 paper
- ▶ Hugging Face documentation

- ▶ **Examples:**

- ▶ BloombergGPT (financial LLM)
- ▶ GitHub Copilot (code generation)
- ▶ Google Search with BERT